1. If $y = \sqrt{\sin(\csc \pi x)}$, then the product $2yy'$ is equal to

$y' = \frac{1}{2\sqrt{\sin(\csc \pi x)}} \cdot \cos(\csc \pi x) \cdot [-\pi \csc(\pi x) \cot(\pi x)] \Rightarrow$

$2yy' = -\pi \cos(\pi x) \csc(\pi x) \cot(\pi x)$.

(a) $2 \sin \pi x . \cos \pi x$

(b) $-\pi \cos(\csc \pi x) . \csc \pi x . \cot \pi x$

(c) $-\pi$

(d) $\cos(\csc \pi x . \cot \pi x)$

(e) π

2. The limit $\lim_{x \to 2} \frac{\tan(x-2)}{4x^2 - 8x} = \lim_{x \to 2} \frac{\tan(x-2)}{4x(x-2)} = I$

Let $\theta = x - 2 \Rightarrow x \to 2$ then $\theta \to 0$

$\Rightarrow I = \lim_{\theta \to 0} \frac{\tan \theta}{4(\theta+2)\theta}$

$= \lim_{\theta \to 0} \left(\frac{\tan \theta}{\theta}\right)\left(\frac{1}{4(\theta+2)}\right)$

$= 1 \cdot \frac{1}{8} = \frac{1}{8}$

(a) is equal to 1

(b) is equal to $\frac{1}{4}$

(c) is $-\infty$

(d) is ∞

(e) is equal to $\frac{1}{8}$

3. If $y = (\sqrt{x} - \frac{1}{\sqrt{x}})^2$, then $y'' =$

$y = x - 2 + \frac{1}{x} = x - 2 + x^{-1} \Rightarrow$

$y' = 1 - x^{-2} \Rightarrow y'' = 2x^{-3}$

(a) $-2\sqrt{x^3}$

(b) $4x^{-3}$

(c) $2x^{-3}$

(d) $2\sqrt{x^3}$

(e) $-3x^{-2}$

4. A particle moves along the curve $xy + y^2 = 35$. As it reaches the point $(2, 5)$ the $x-$coordinate is increasing at a rate of $2\,cm/sec.$ How fast is the y-coordinate changing at that instant?

Given $\left.\frac{dx}{dt}\right|_{(2,5)} = 2$ cm/sec Need $\left.\frac{dy}{dt}\right|_{(2,5)}$.

(a) $-3\,cm/sec.$

(b) $3\,cm/sec.$

(c) $-\frac{7}{6}\,cm/sec.$

(d) $-\frac{5}{6}\,cm/sec.$

(e) $\frac{1}{6}\,cm/sec.$

$\frac{d}{dt}(xy + y^2) = \frac{d}{dt}(35) \Rightarrow$

$x\frac{dy}{dt} + y\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$

at $(2,5) \Rightarrow$

$2\frac{dy}{dt} + (5)(2) + 2(5)\frac{dy}{dt} = 0$

$\Rightarrow 12\frac{dy}{dt} = -10 \Rightarrow \frac{dy}{dt} = -\frac{5}{6}$ cm/sec.

5. If $y = \dfrac{\tan x}{1 + x \tan x}$, then $y' = \dfrac{g(x)}{(1 + x \tan x)^2}$ where $g(x) =$

$$y' = \frac{(1+x\tan x)\sec^2 x - \tan x(\tan x + x\sec^2 x)}{(1+x\tan x)^2}$$

(a) $\sec x$

(b) 1

(c) -2

(d) $\sec x \tan x$

(e) $\tan x$

$$= \frac{\sec^2 x - \tan^2 x}{(1+x\tan x)^2} = \frac{1}{(1+x\tan x)^2}$$

$$\Rightarrow \quad g(x) = 1$$

6. The slope of the tangent line to the graph of $y = (1 + 4x^2)\tan^{-1}(2x)$ at $x = \dfrac{1}{2}$ is

(a) $2 + \pi$

(b) $2 - \pi$

(c) $8 + 2\pi$

(d) $1 + 2\pi$

(e) $4 + \pi$

$$y' = (1+4x^2)\frac{2}{(1+4x^2)} + 8x\tan^{-1}(2x)$$

$$y'\Big|_{x=1/2} = 2 + 4\tan^{-1}1 = 2 + 4\left(\frac{\pi}{4}\right) = 2 + \pi$$

7. If $f(x) = 2^x \cdot x^2$, then $f'(2) =$

(a) $8(2+\ln 2)$

(b) $\ln 16 + ln2$

(c) $16(\ln 2 - 1)$

(d) $4(1+\ln 2)$

~~(e)~~ $16(1+\ln 2)$

$f'(x) = 2^x(2x) + x^2(2^x \ln 2) \Rightarrow$

$f'(2) = (4)(4) + (4)(4)\ln 2$

$= 16 + 16 \ln 2$

$= 16(1 + \ln 2)$

8. If $f'(x) = \lim_{h\to 0} \dfrac{e^{2x}(e^{2h}-1)}{3h}$, then a possible expression for the function f is

~~(a)~~ $f(x) = \frac{1}{3}e^{2x}$

(b) $f(x) = 3(e^{2x}-1)$

(c) $f(x) = 3e^{2x}$

(d) $f(x) = \frac{1}{3}(e^x+1)^2$

(e) $f(x) = e^{2x} + \frac{1}{3}$

$f'(x) = \frac{1}{3}\lim_{h\to 0} \dfrac{e^{2(x+h)} - e^{2x}}{h}$

$= \frac{1}{3}\frac{d}{dx}(e^{2x})$

$= \frac{d}{dx}\left(\frac{1}{3}e^{2x}\right)$

Thus $f(x) = \frac{1}{3}e^{2x}$ is a possible expression for f.

9. If $f(x) = x^{\ln x}$, then $f'(e^2) =$

(a) $\frac{1}{2}e^2$

(b) $\frac{1}{4}e^2$

(c) $2e^2$

(d) $4e^2$

(e) e^2

$$\ln f(x) = (\ln x)(\ln x) = (\ln x)^2 \Rightarrow$$

$$\frac{f'(x)}{f(x)} = 2(\ln x)\cdot\left(\frac{1}{x}\right) \Rightarrow$$

$$f'(x) = \frac{2\, x^{\ln x} \ln x}{x} \Rightarrow$$

$$f'(e^2) = \frac{2\, e^{2\ln e^2} \ln e^2}{e^2} \Rightarrow$$

$$f'(e^2) = \frac{2}{e^2}\left(e^{\ln e^4}\right)\cdot(2) = \frac{4}{e^2} e^4 = 4e^2 .$$

10. If a particle is moving according to the law of motion $S(t) = 3\cos\left(\frac{\pi}{2}t\right)$ where t is measured in seconds and S in meters, then the total distance traveled by the particle during the time interval $[0, 5]$ is

(a) 15 meters

(b) 6 meters

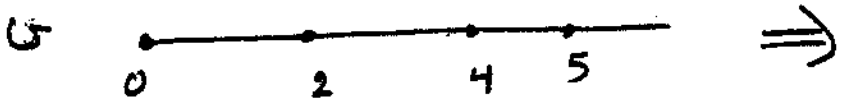
(c) 12 meters

(d) 9 meters

(e) 18 meters

$$v = S'(t) = -3\left(\frac{\pi}{2}\right)\sin\frac{\pi}{2}t$$

$\sin\frac{\pi}{2}t = 0$ when $t = 0, 2, 4$ on $[0, 5]$



$\Rightarrow$

The total distance required:

$$S = |S(2) - S(0)| + |S(4) - S(2)| + |S(5) - S(4)|$$

$$= |-3 - 3| + |3 - (-3)| + |0 - 3|$$

$$= 6 + 6 + 3 = 15 \text{ meters}$$

11. If $y = \ln\left[\frac{2}{3}\left(x+\sqrt{x^2-1}\right)\right]$, then $\frac{dy}{dx} =$

(a) $\frac{2}{3\sqrt{x^2-1}}$

(b) $\frac{2}{3}\left(1+x\sqrt{x^2-1}\right)$

(c) $\frac{1}{x+\sqrt{x^2-1}}$

(d) $\frac{3}{2\sqrt{x^2-1}}$

(e) $\frac{1}{\sqrt{x^2-1}}$

$$y' = \frac{\frac{2}{3}\left(1+\frac{2x}{2\sqrt{x^2-1}}\right)}{\frac{2}{3}\left(x+\sqrt{x^2-1}\right)} = \frac{\left(\frac{\sqrt{x^2-1}+x}{\sqrt{x^2-1}}\right)}{x+\sqrt{x^2-1}} = \frac{1}{\sqrt{x^2-1}}.$$

12. The volume of a sphere is changing at the rate of $4\pi\ cm^3/sec.$ How fast is the diameter of the sphere changing when its volume is $\frac{32\pi}{3}\ cm^3$? [Volume of a sphere $= \frac{4\pi}{3}(radius)^3$].

(a) $\frac{3}{4}\ cm/sec.$

(b) $\frac{3}{2}\ cm/sec.$

(c) $\frac{3}{8}\ cm/sec.$

(d) $\frac{9}{8}\ cm/sec.$

(e) $\frac{1}{2}\ cm/sec.$

If R is the diameter $\Rightarrow$

Volume $V = \frac{4\pi}{3}\left(\frac{R}{2}\right)^3 = \frac{\pi R^3}{6} \Rightarrow$

$\frac{dV}{dt} = \frac{\pi}{2}R^2\frac{dR}{dt} \rightarrow (1)$

When $V = \frac{32\pi}{3} \Rightarrow \frac{32\pi}{3} = \frac{\pi R^3}{6} \Rightarrow$

$R^3 = 64 \Rightarrow R = 4 \rightarrow (2)$

$\Rightarrow$ (1) & (2) and $\frac{dV}{dt} = 4\pi\ cm^3/sec \Rightarrow$

$4\pi = \frac{\pi}{2}(16)\frac{dR}{dt} \Rightarrow \frac{dR}{dt} = \frac{1}{2}\ cm/sec.$

13. If $(0, \alpha)$ are the coordinates of the y-intercept of the tangent line to the graph of $y = \sin^{-1}(2x)$ at $x = \frac{\sqrt{2}}{4}$, then $\alpha =$

(a) $\frac{\pi}{4} - \sqrt{2}$

(b) $\frac{\pi}{4} + \sqrt{2}$

(c) $\frac{\pi}{4} - 1$

(d) $\frac{\pi}{4} - \frac{1}{4}$

(e) $\pi\sqrt{2}$

$y' = \frac{2}{\sqrt{1-4x^2}} \Rightarrow y'\left(\frac{\sqrt{2}}{4}\right) = \frac{2}{\sqrt{\frac{1}{2}}} = 2\sqrt{2}$.

at $x = \frac{\sqrt{2}}{4} \Rightarrow y = \sin^{-1}\left(\frac{\sqrt{2}}{2}\right) = \frac{\pi}{4} \Rightarrow$

slope $= 2\sqrt{2}$, point $\left(\frac{\sqrt{2}}{4}, \frac{\pi}{4}\right) \Rightarrow$

eqn of tangent is

$y - \frac{\pi}{4} = 2\sqrt{2}\left(x - \frac{\sqrt{2}}{4}\right)$

at y-intercept we have $x = 0$, $y = \alpha$

$\Rightarrow \alpha - \frac{\pi}{4} = 2\sqrt{2}\left(0 - \frac{\sqrt{2}}{4}\right) = -1 \Rightarrow$

$\alpha = \frac{\pi}{4} - 1$

14. The slope of the normal line to the curve $e^{y/x} = x - y$ at the point $(1, 0)$ is

(a) -1

(b) $\frac{1}{3}$

(c) -2

(d) -3

(e) 1

we use implicit differentiation to find y':

$\frac{d}{dx}\left(e^{y/x}\right) = \frac{d}{dx}(x - y) \Rightarrow$

$e^{y/x}\left(\frac{xy' - y}{x^2}\right) = 1 - y'$

at $(1, 0)$ i.e $x = 1$ and $y = 0 \Rightarrow$

$(1)\left(\frac{y' - 0}{1}\right) = 1 - y' \Rightarrow y' = 1 - y'$

$\Rightarrow 2y' = 1 \Rightarrow y' = \frac{1}{2} \Rightarrow$

The slope of the normal line at $(1, 0) = -2$

15. If the tangent line to the graph of $y = x^2 - 4x + 19$ at the point $P(\alpha, \beta)$, where $\alpha < 0$, passes through the point $(3, 0)$, then $\beta =$

(α,β) is on the curve $\Rightarrow \boxed{\beta = \alpha^2 - 4\alpha + 19}$ ①

(a) 30

(b) 18

(c) 28

(d) 36

~~(e) 24~~

The slope of the graph at any point is $y' = 2x - 4 \Rightarrow$ The slope at $(\alpha,\beta) = 2\alpha - 4$ ⊛

But the tangent line passes through (α,β) and $(3,0) \Rightarrow$ its slope also can be obtained by $\frac{y_1 - y_2}{x_1 - x_2} = \frac{\beta - 0}{\alpha - 3}$ ⊛⊛ Now ⊛ ≡ ⊛⊛ $\Rightarrow$

$2\alpha - 4 = \frac{\beta}{\alpha - 3} \Rightarrow \boxed{\beta = 2\alpha^2 - 10\alpha + 12}$ ② ①+② $\Rightarrow$

$2\alpha^2 - 10\alpha + 12 = \alpha^2 - 4\alpha + 19 \Rightarrow \alpha^2 - 6\alpha - 7 = 0 = (\alpha+1)(\alpha-7) \Rightarrow$

$\alpha = -1, \alpha \neq 7 \Rightarrow \beta = \alpha^2 - 4\alpha + 19 = 1 + 4 + 19 = 24$

16. Which one of the following statements is TRUE?

(a) If f is continuous at a, then f is differentiable at a (FALSE)

(b) If $f(a) = 0$ and $f'(a) = 0$, then $f''(a) = 0$ (FALSE)

(c) If $f'(a)$ does not exist, then f is discontinuous at a (FALSE)

(d) If f' is differentiable at a, then $f''(a)$ exists (FALSE)

(e) If $\lim_{x \to a} f'(x)$ exists, then f' is continuous at a (TRUE)

(Ask Yourself why?)

In case a false statement give a simple example

for example: (a) is FALSE because $f(x) = |x|$ is continuous at $x = 0$ but not differentiable at $x = 0$

Now you try the rest !!!!

17. If $f(x) = xe^{-x}$, then $f^{(100)}(x) = (Ax+B)e^{-x}$, where A and B are constants and $A+B=$

we get $f^{(100)}(x)$ by induction: $f'(x) = -xe^{-x} + e^{-x} \Rightarrow$

(a) 99 $f'(x) = (-x+1)e^{-x} \Rightarrow f''(x) = -(-x+1)e^{-x} - e^{-x} \Rightarrow$

~~(b)~~ -99 $f''(x) = (x-2)e^{-x} \Rightarrow f'''(x) = -(x-2)e^{-x} + e^{-x} \Rightarrow$

(c) 101 $f'''(x) = (-x+3)e^{-x} \Rightarrow f^{(4)}(x) = -(-x+3)e^{-x} - e^{-x} \Rightarrow$

(d) -2 $f^{(4)}(x) = (x-4)e^{-x}$. Thus by induction (observe

(e) -100 that 100 is even) $\Rightarrow$

$f^{(100)}(x) = (x-100)e^{-x} \Rightarrow A=1,\ B=-100$

$\Rightarrow A+B = -99$

18. $\lim_{x\to 0} \dfrac{5x^2}{2x - 2x\cos x + 2\sin^2 3x}$ (0/0 type!)

(a) is equal to 0

(b) is equal to 5

~~(c)~~ is equal to $\frac{5}{18}$

(d) does not exist

(e) is equal to $\frac{5}{3}$

$$= \lim_{x\to 0} \frac{5}{\frac{2x(1-\cos x)}{x^2} + 2\left(\frac{\sin 3x}{x}\right)^2}$$

$$= \lim_{x\to 0} \frac{5}{2\left(\frac{1-\cos x}{x}\right) + 2\left(\frac{\sin 3x}{x}\right)^2}$$

$$= \lim_{x\to 0} \frac{5}{2(0) + 2(3)^2} = \frac{5}{18}$$

19. Given the function $f(x) = \begin{cases} x+3 & \text{if } x < -2 \\ |x+1| & \text{if } -2 \le x \le 8. \\ (x-6)^2 & \text{if } x > 8 \end{cases}$

The sum of all values of x for which the function f is not differentiable is

$$f(x) = \begin{cases} x+3 & \text{if } x < -2 \\ -(x+1) & \text{if } -2 \le x \le -1 \\ x+1 & \text{if } -1 < x \le 8 \\ (x-6)^2 & \text{if } x > 8 \end{cases} \Rightarrow$$

(a) 0

(b) 6

(c) 4

(d) 3

(e) 5

$f'_-(-2) = 1 \neq f'_+(-2) = -1$ (See #54 Page 165)

$f'_-(-1) = -1 \neq f'_+(-1) = 1$

$f'_-(8) = 1 \neq f'_+(8) = 2(8-6) = 4$

$\Rightarrow$ f is not differentiable at $-2, -1$ and 8

$\Rightarrow -2-1+8 = 5$

20. Which one of the following statements is TRUE about the function $f(x) = \sqrt[3]{x}$?

$f'(x) = \frac{1}{3}x^{-2/3} = \frac{1}{3\sqrt[3]{x^2}} \Rightarrow$

NO (a) $\lim_{x \to 0^-} f'(x) = -\infty$ $\lim_{x \to 0} |f'(x)| = \infty$ and f is continuous at $x = 0$

NO (b) The graph of f has a vertical asymptote at $x = 0$

Hence (c) is the correct answer.

(c) The graph of f has vertical tangent at $x = 0$

NO (d) The graph of f' lies below the x-axis on $(-\infty, 0)$

NO (e) f is differentiable at $x = 0$

because $f' > 0$ for all $x \neq 0$.